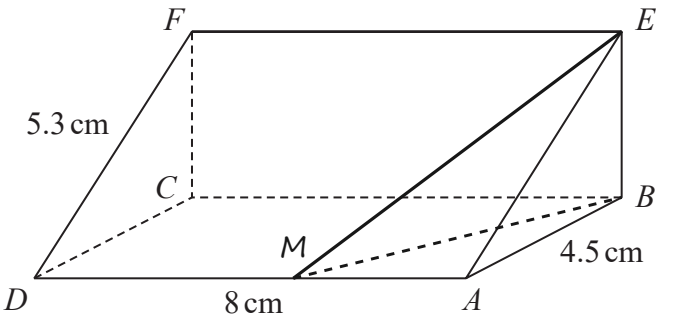
20 The diagram shows a triangular prism.



The base, *ABCD*, of the prism is a rectangle.
Angle *DCF* and angle *ABE* are right angles.

M is the point on *DA* such that *DM* : *MA* = 3 : 2

Calculate the size of the angle between *EM* and the base of the prism.
Give your answer correct to 1 decimal place.

$BE^2 = 5.3^2 - 4.5^2 = 7.84$, so $BE = 2.8$ cm

1 mark
Height; ratio along DA.

$MA = 8 \div 5 \times 2 = 3.2$ cm

$MB^2 = 4.5^2 + 3.2^2 = 30.49$

1 mark
Pythagoras in the base.

$MB = \sqrt{30.49} = 5.521775077...$ cm

Required angle is EMB: tan angle = BE ÷ MB

1 mark
Use right triangle EMB.

Angle = $\tan^{-1}(2.8 \div \sqrt{30.49}) = 26.88880557...°$

= 26.9° (1 decimal place)

1 mark
26.9°.

°

(Total for Question 20 is 4 marks)

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